

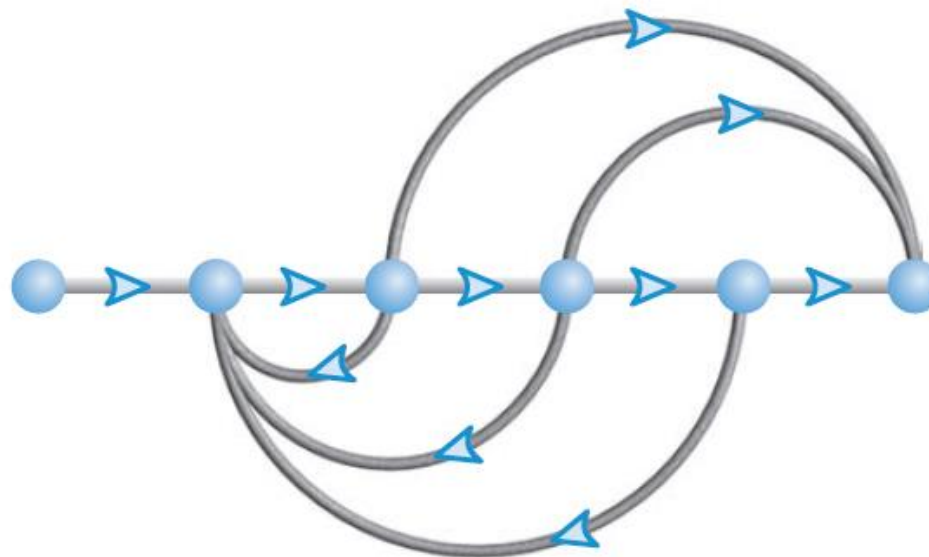


Artificial Intelligence Engineering Department



LECTURES (5-6)

REDUCTION OF MULTIPLE SUBSYSTEMS



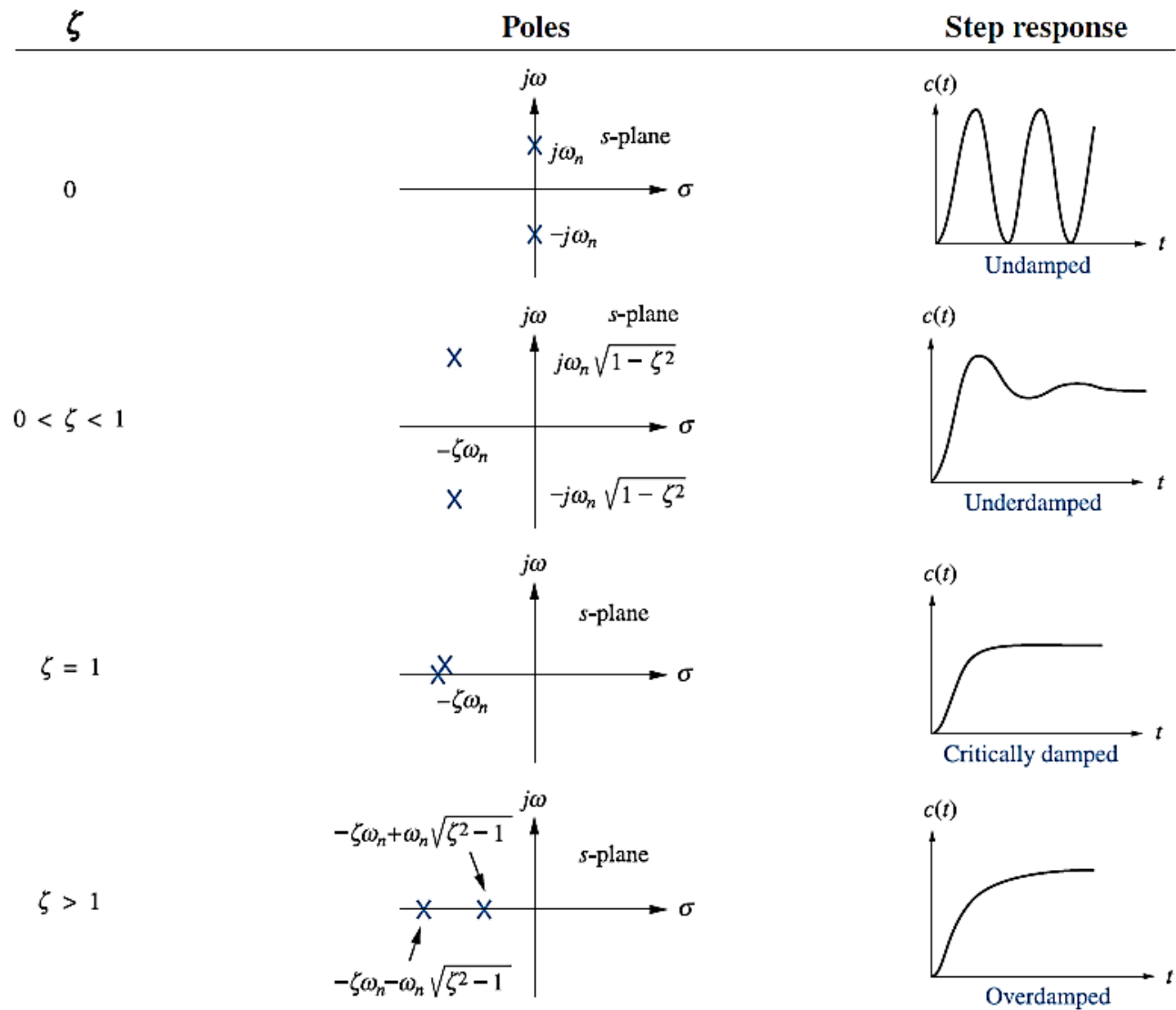
Outline

- **Step response -Second-Order Prototype System**
- **Block diagrams**
- **Analysis and Design of Feedback Systems**
- **Signal Flow Graph**



$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

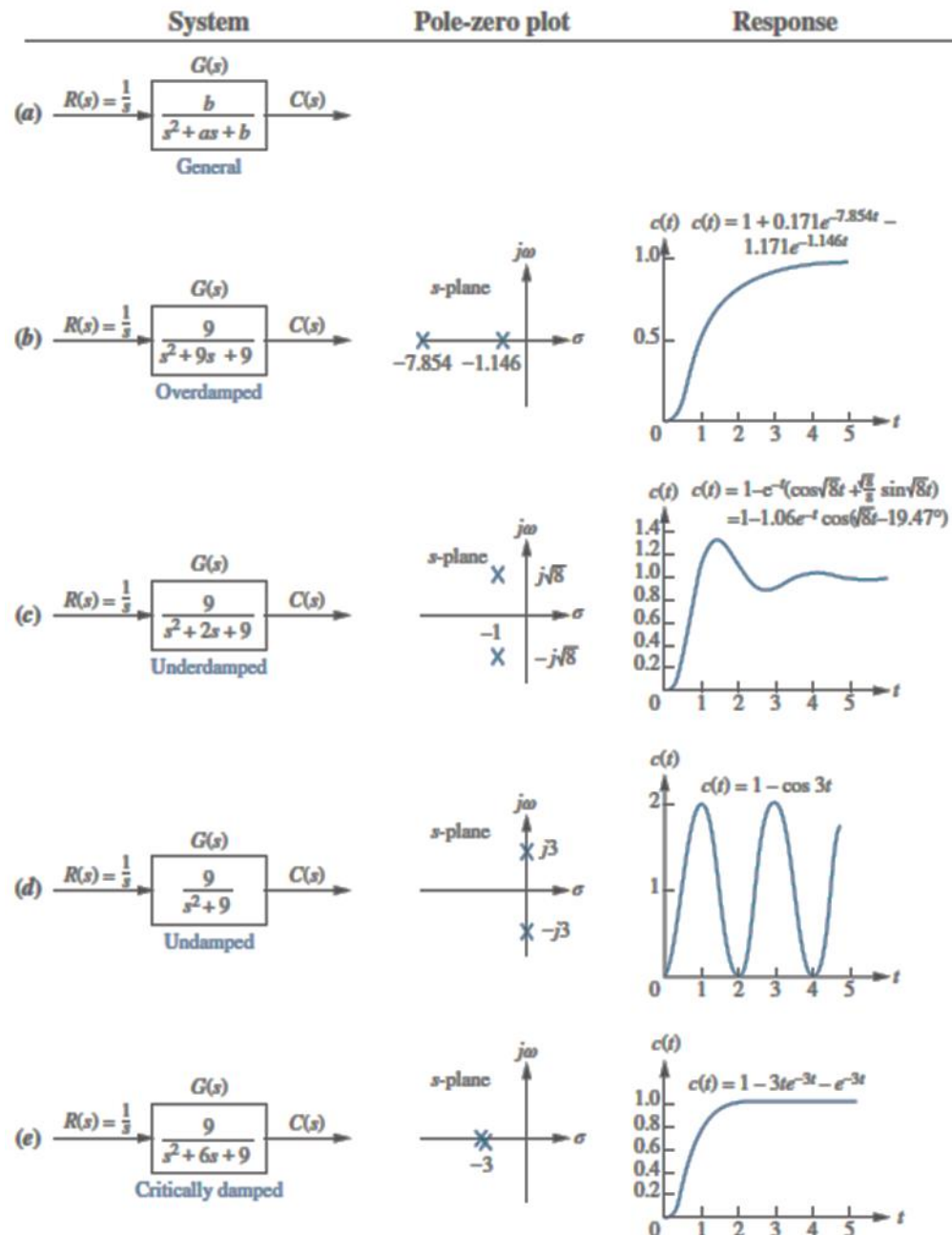
$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$



Examples:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

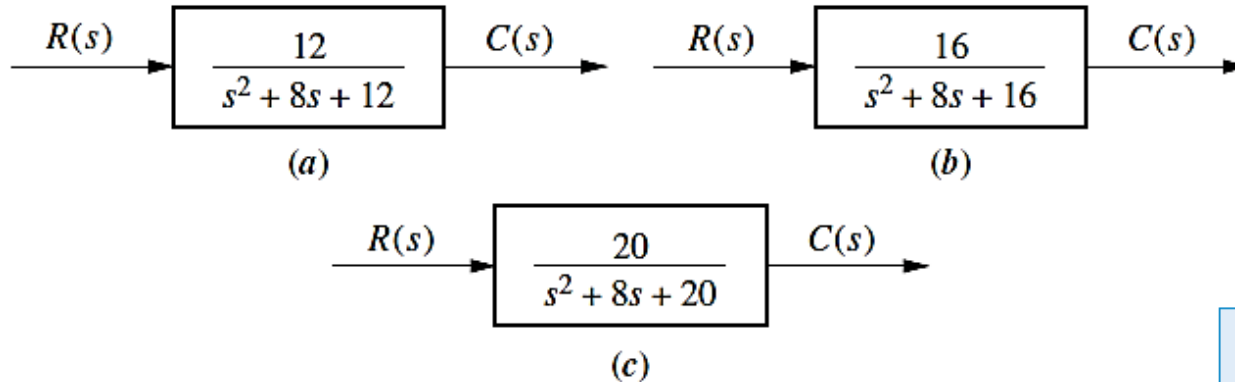


PROBLEM: Given the transfer function of Eq. (4.23), find ζ and ω_n .

$$G(s) = \frac{36}{s^2 + 4.2s + 36} \quad \longleftrightarrow \quad G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

SOLUTION: Comparing With standard Eq , $\omega_n^2 = 36$, from which $\omega_n = 6$. Also, $2\zeta\omega_n = 4.2$. Substituting the value of ω_n , $\zeta = 0.35$.

PROBLEM: For each of the systems shown in Figure find the value of ζ and report the kind of response expected.



$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

SOLUTION: First match the form of these systems to the forms shown in Eq

Since $a = 2\zeta\omega_n$ and $\omega_n = \sqrt{b}$,

$$\zeta = \frac{a}{2\sqrt{b}}$$

$\zeta = 1.155$ for system a (overdamped $\zeta > 1$)

$\zeta = 1$ for system b (critically damped $\zeta = 1$)

$\zeta = 0.894$ for system c (underdamped $\zeta < 1$)

PROBLEM:

For each of the transfer functions do the following: (1) Find the values of ζ and ω_n ; (2) characterize the nature of the response.

a. $G(s) = \frac{400}{s^2 + 12s + 400}$

b. $G(s) = \frac{900}{s^2 + 90s + 900}$

c. $G(s) = \frac{225}{s^2 + 30s + 225}$

d. $G(s) = \frac{625}{s^2 + 625}$

ANSWERS:

a. $\zeta = 0.3$, $\omega_n = 20$; system is underdamped

b. $\zeta = 1.5$, $\omega_n = 30$; system is overdamped

c. $\zeta = 1$, $\omega_n = 15$; system is critically damped

d. $\zeta = 0$, $\omega_n = 25$; system is undamped

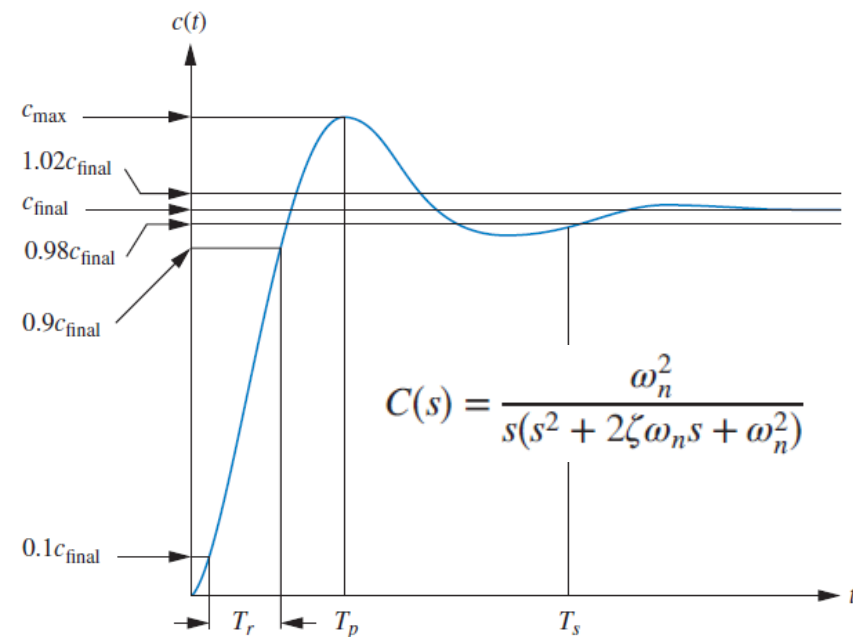
• **Underdamped Second-Order Systems:**

1- Rise time, T_r . The time required for the waveform to go from 0.1 of the final value to 0.9 of the final value.

2- Peak time, T_p . The time required to reach the first, or maximum, peak.

3-Percent overshoot, %OS. The amount that the waveform overshoots the steady-state, or final, value at the peak time.

4-Settling time, T_s . The time required for the transient's damped oscillations to reach and stay within 2% of the steady-state value



• Underdamped Second-Order Systems:

1- Evaluation of T_p :

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

2- Evaluation of %OS :

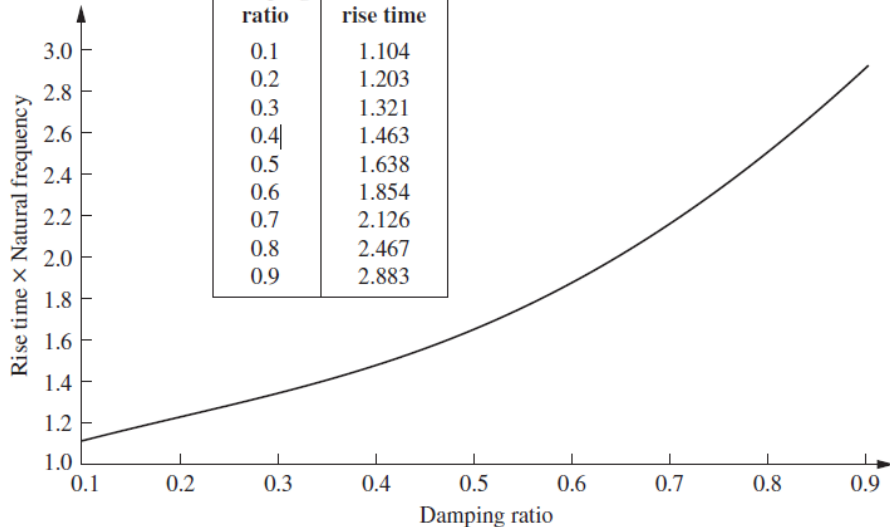
$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100$$

3- Evaluation of T_s :

$$T_s = \frac{4}{\zeta\omega_n}$$

4- Evaluation of T_r :

Damping ratio	Normalized rise time
0.1	1.104
0.2	1.203
0.3	1.321
0.4	1.463
0.5	1.638
0.6	1.854
0.7	2.126
0.8	2.467
0.9	2.883



PROBLEM: Given the transfer function

$$G(s) = \frac{100}{s^2 + 15s + 100}$$

find T_p , %OS, T_s , and T_r .

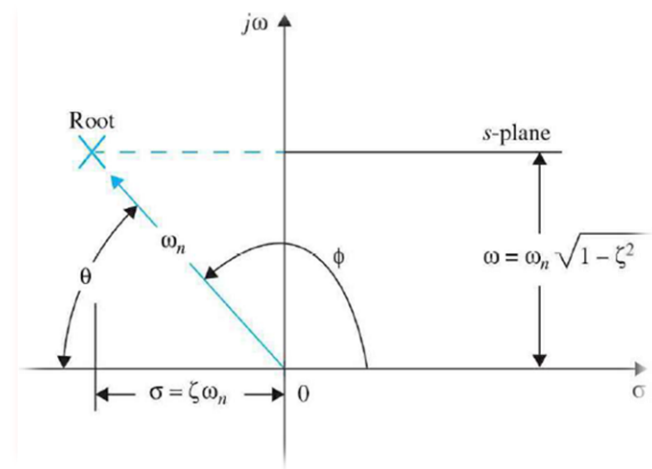
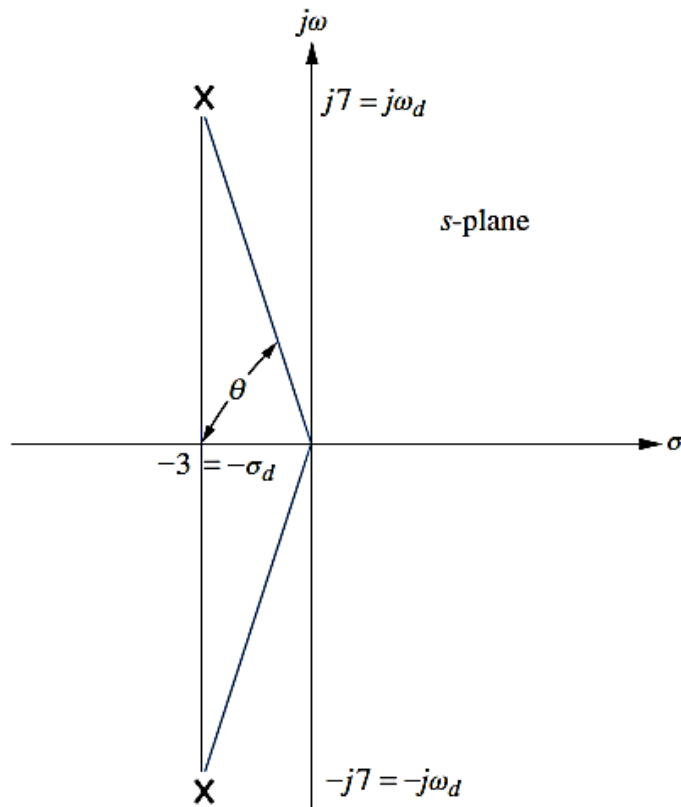
SOLUTION:

- $\omega_n = 10$
- $\zeta = 0.75$
- $T_p \cong 0.475$ second
- %OS=2.838,
- $T_s = 0.533$ second
- Using the table in Figure, the normalized rise time is approximately 2.3 seconds.
- Dividing by ω_n yields :
 $T_r \cong 0.23$ second.



• Underdamped Second-Order Systems:

PROBLEM: Given the pole plot shown in Figure find ζ , ω_n , T_p , %OS, and T_s .



SOLUTION: The damping ratio is given by $\zeta = \cos \theta = \cos[\arctan(7/3)] = 0.394$. The natural frequency, ω_n , is the radial distance from the origin to the pole, or $\omega_n = \sqrt{7^2 + 3^2} = 7.616$. The peak time is

$$T_p = \frac{\pi}{\omega_d} = \frac{\pi}{7} = 0.449 \text{ second}$$

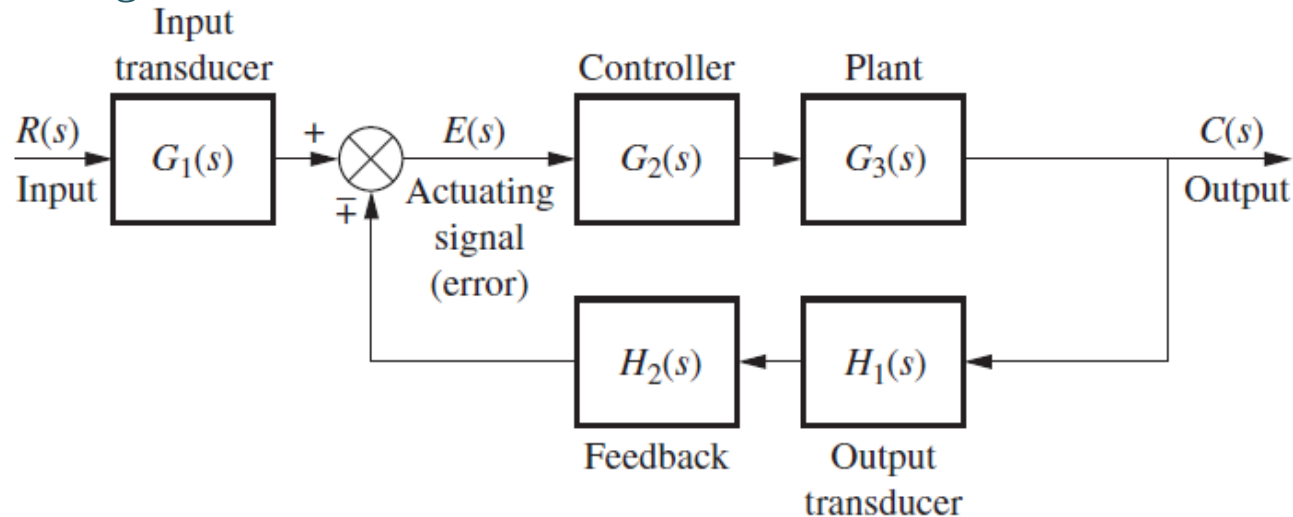
The percent overshoot is

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 = 26\%$$

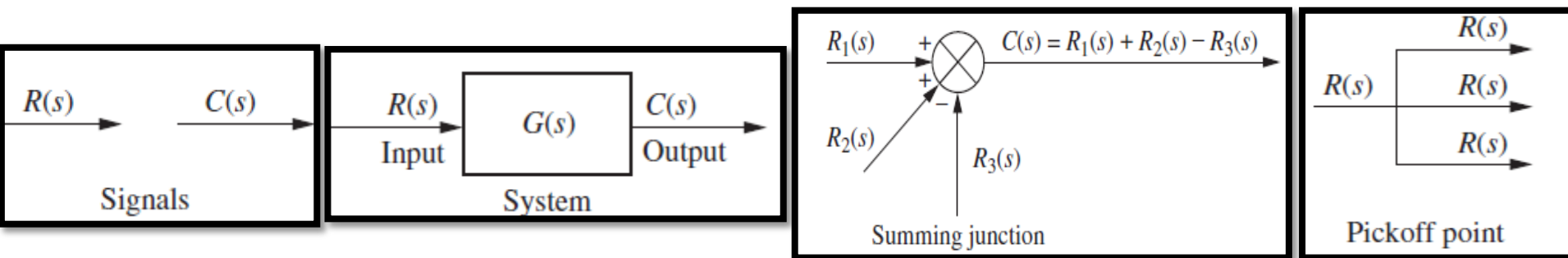
The approximate settling time is

$$T_s = \frac{4}{\sigma_d} = \frac{4}{3} = 1.333 \text{ seconds}$$

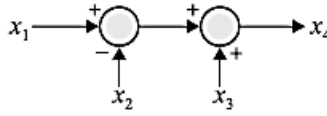
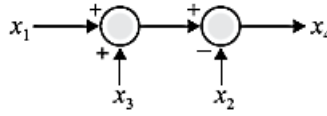
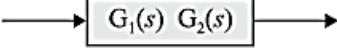
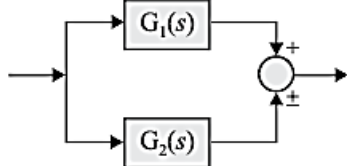
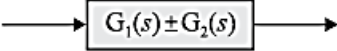
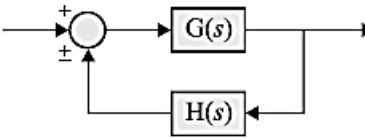
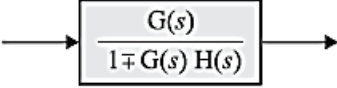
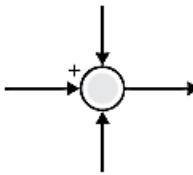
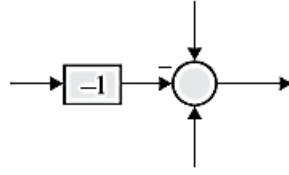
- Many systems are composed of multiple subsystems.
- When multiple subsystems are interconnected, a few more schematic elements must be added to the block diagram.
- A control system usually consists of a number of components and multiple feedback loops. As following Ex



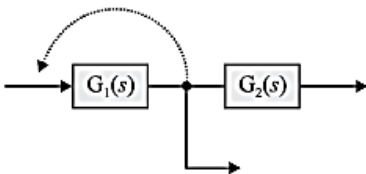
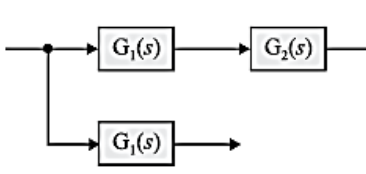
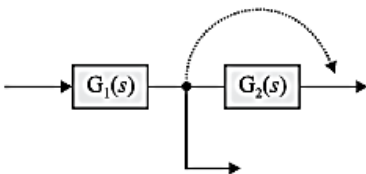
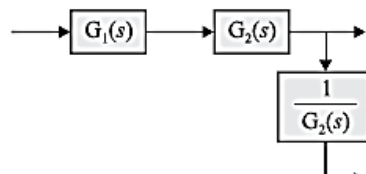
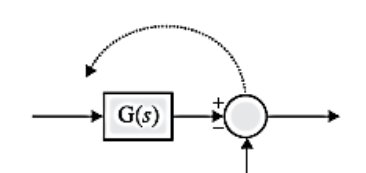
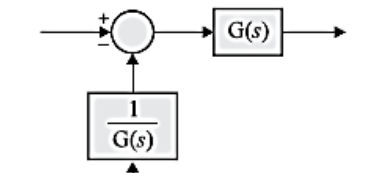
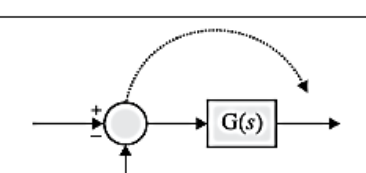
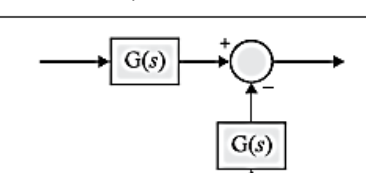
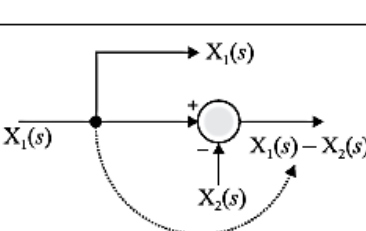
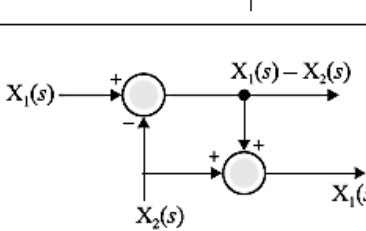
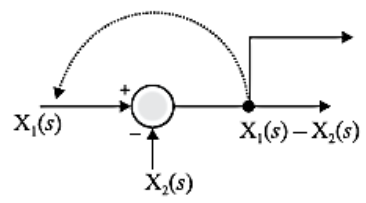
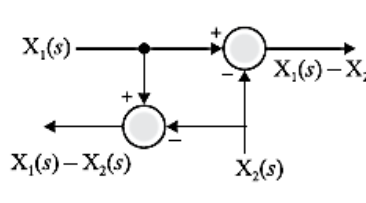
- A block diagram has following basic elements:



- Block diagram reduction means simplifying a diagram consisting of many blocks and feedback loops to the extent that it is a single block providing a transfer function that relates the output to the input. The rules that are followed in doing so are listed in the following Table:

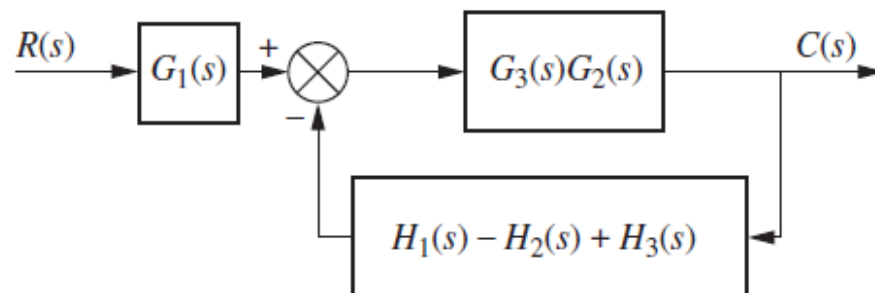
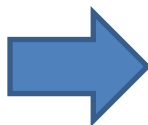
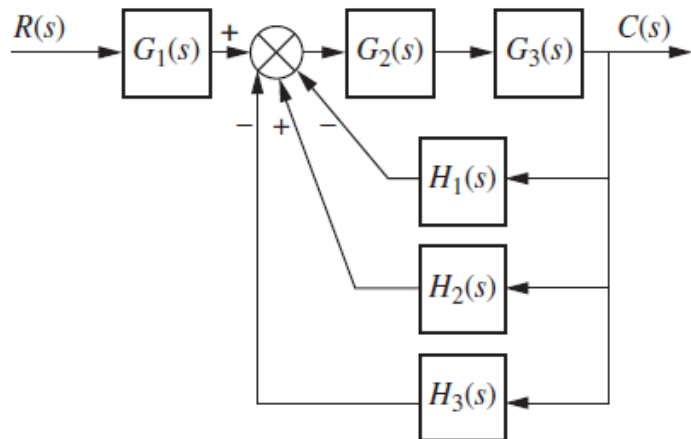
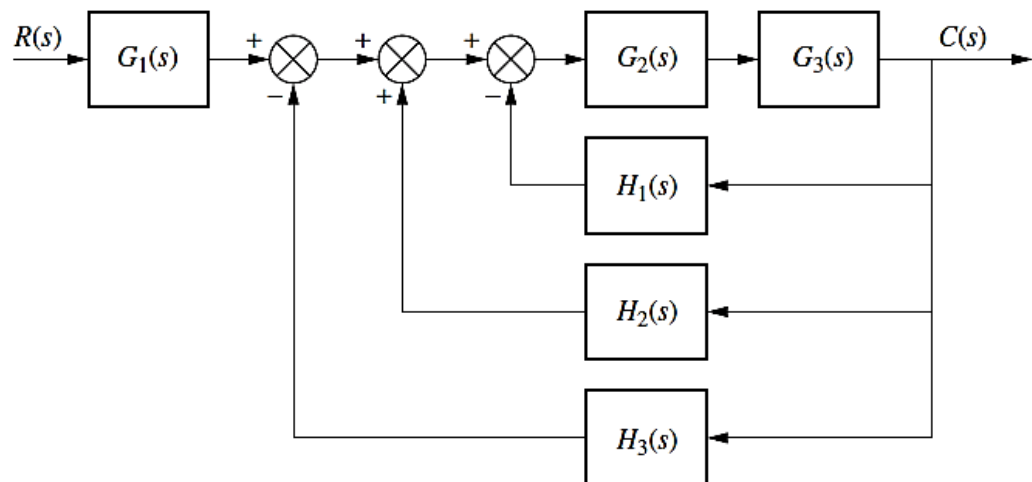
	Rule	Original diagram	Equivalent diagram
1.	Associative law for summing points with no additional block or connection in between		
2.	Blocks in cascade (or series) with no additional connection in between		
3.	Blocks in tandem (or parallel)		
4.	Two blocks in feedback configuration		
5.	Insertion / removal of unity gain		
6.	Changing a summer sign		



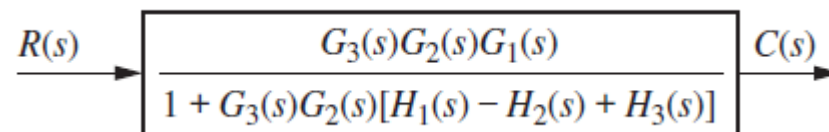
	Rule	Original diagram	Equivalent diagram
7.	Moving a take off point back		
8.	Moving a takeoff point forward		
9.	Moving a summing point back		
10.	Moving a summing point forward		
11.	Moving a take off point forward of a summing point		
12.	Moving a take off point behind a summing point		



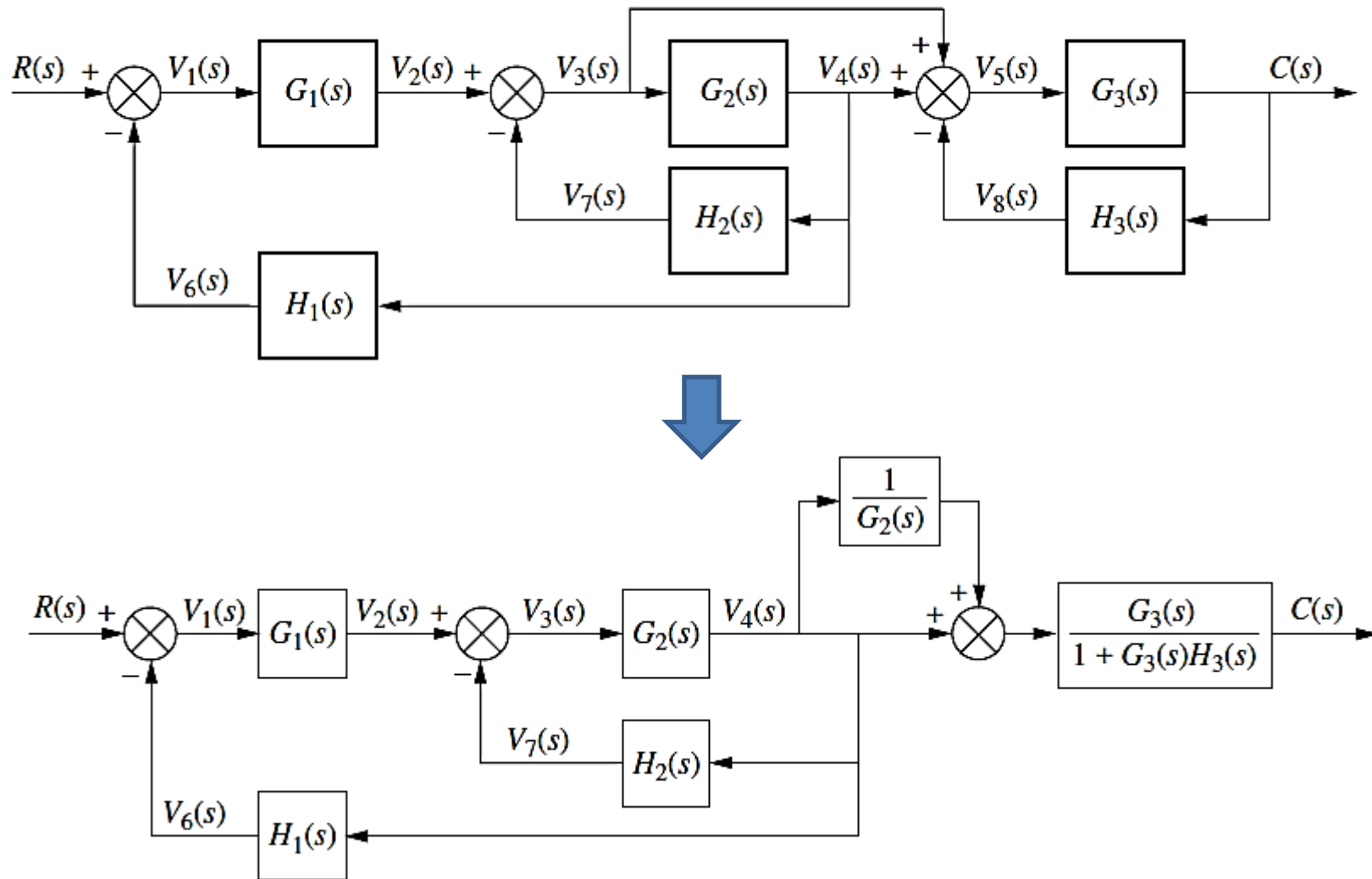
PROBLEM: Reduce the block diagram shown in Figure to a single transfer function.

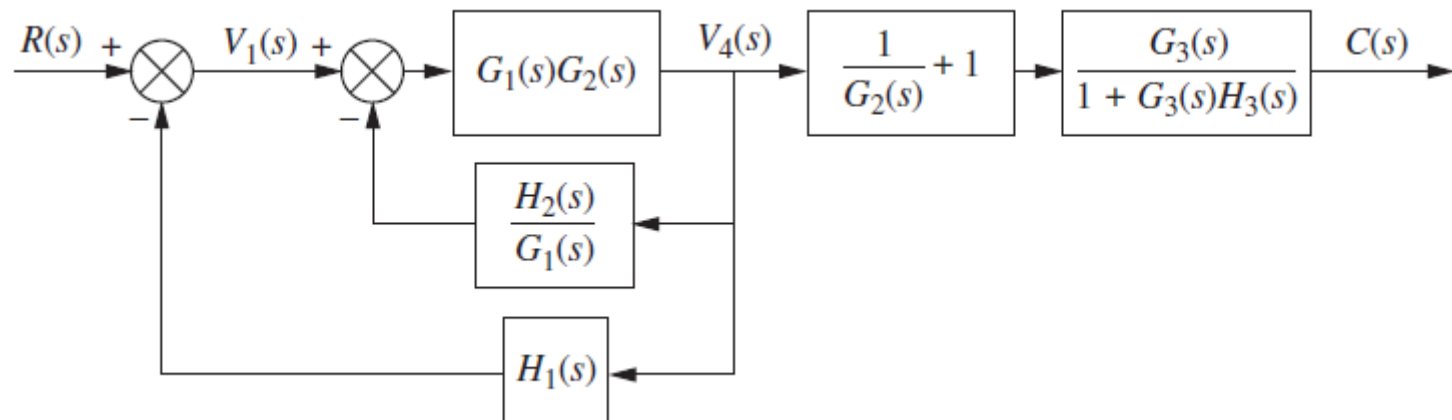
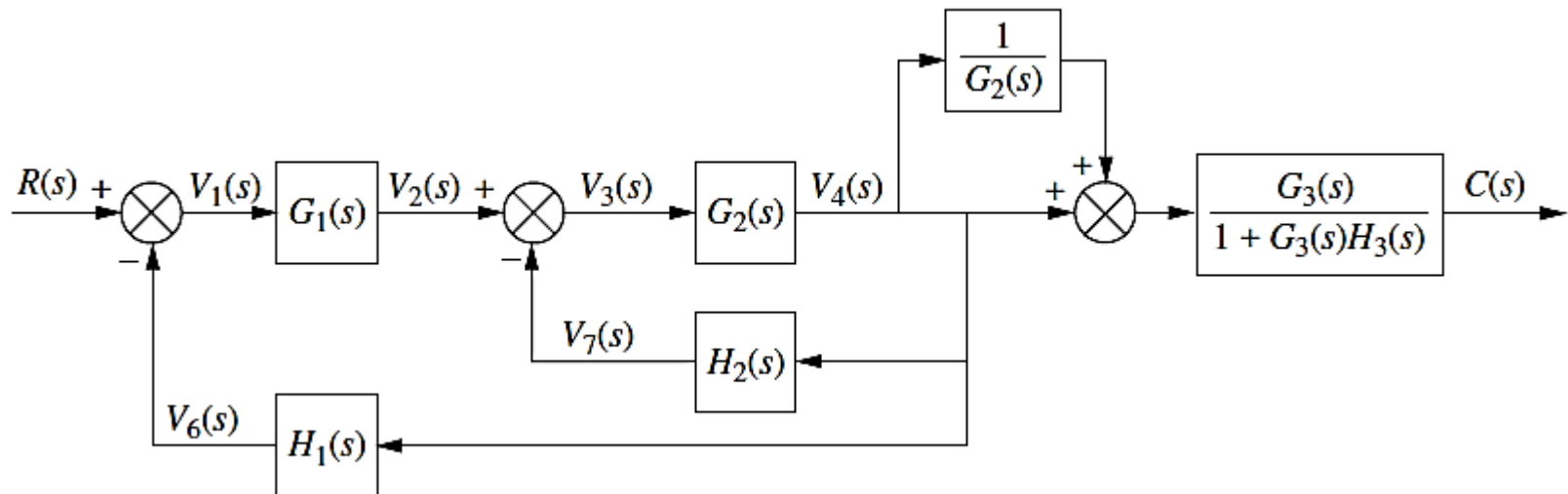


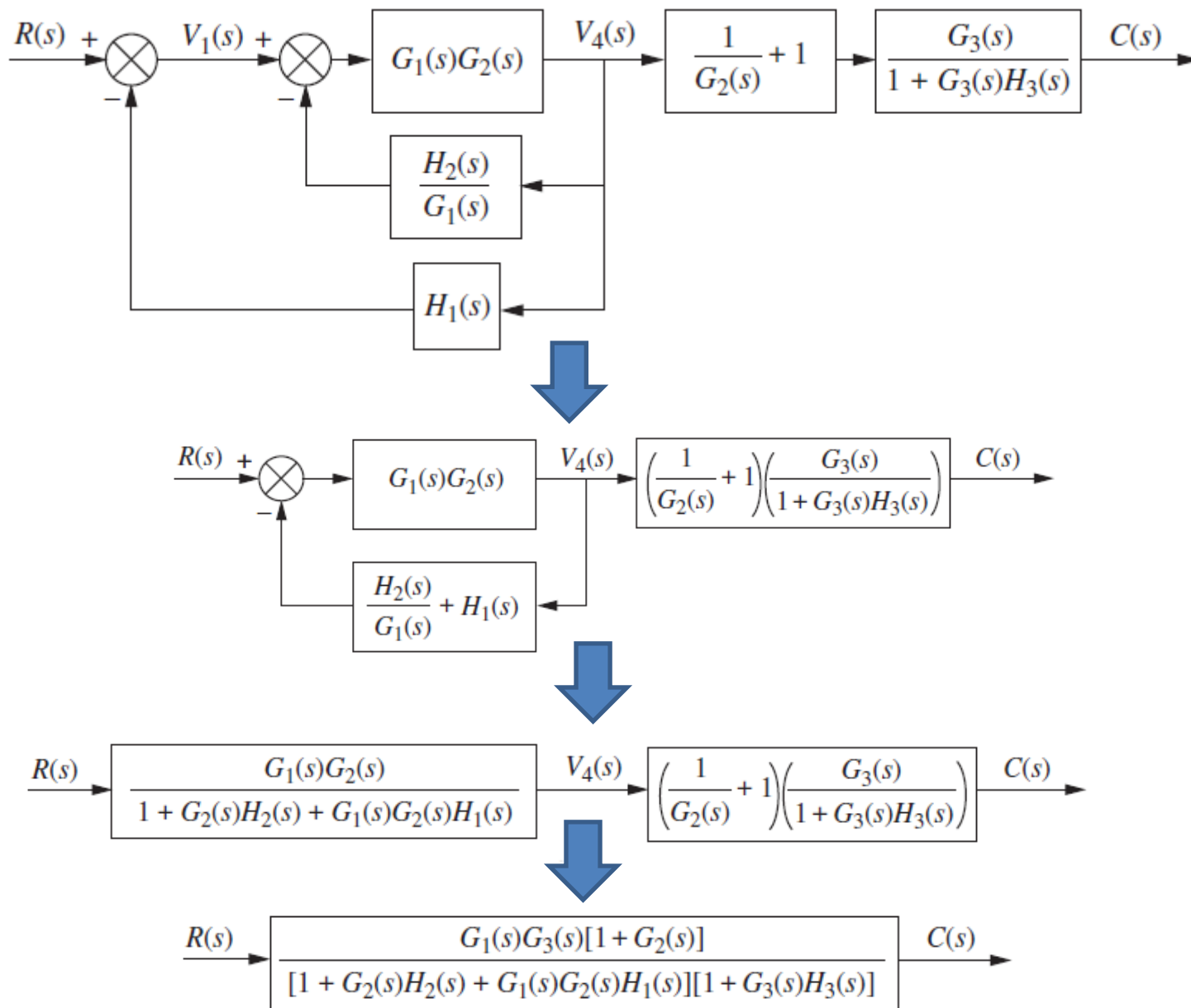
(b)



PROBLEM: Reduce the system shown in Figure to a single transfer function.

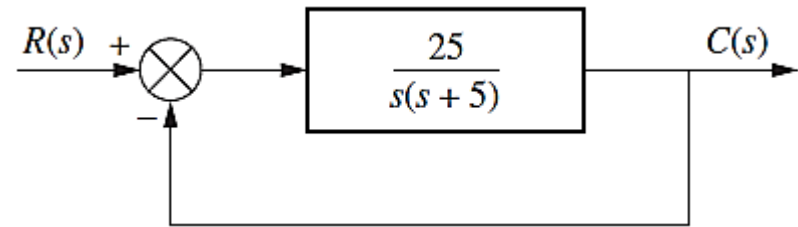






PROBLEM: For the system shown in Figure , find the peak time, percent overshoot, and settling time.

SOLUTION: The closed-loop transfer function



$$T(s) = \frac{25}{s^2 + 5s + 25} \Rightarrow \omega_n = \sqrt{25} = 5 \Rightarrow 2\zeta\omega_n = 5 \Rightarrow \zeta = 0.5$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = 0.726 \text{ second} \Rightarrow \%OS = e^{-\zeta\pi / \sqrt{1 - \zeta^2}} \times 100 = 16.303 \Rightarrow T_s = \frac{4}{\zeta\omega_n} = 1.6 \text{ seconds}$$

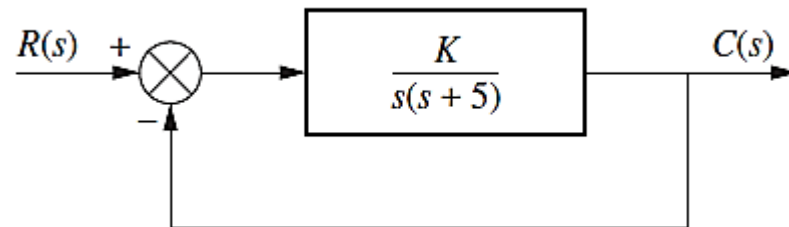
PROBLEM: Design the value of gain. K , for the feedback control system of Figure so that the system will respond with a 10% overshoot.

$$T(s) = \frac{K}{s^2 + 5s + K}$$

$$2\zeta\omega_n = 5$$

$$\omega_n = \sqrt{K}$$

$$\zeta = \frac{5}{2\sqrt{K}}$$

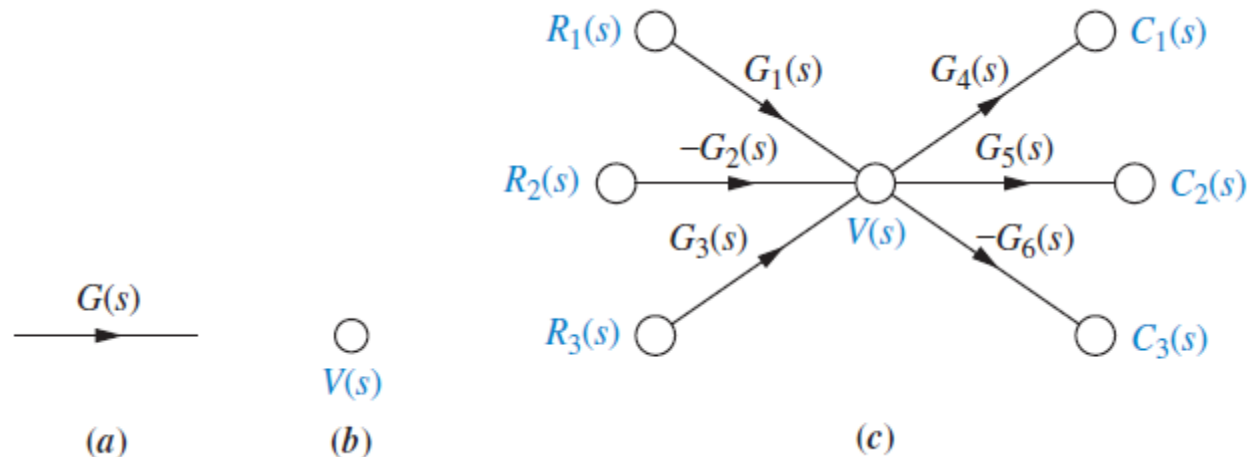


A 10% overshoot implies that $\zeta = 0.591$. Substituting this value for the damping ratio into Eq. and solving for K yields $K = 17.9$



• Signal-Flow Graphs

- Signal-flow graphs are an alternative to block diagrams.
- A signal-flow graph consists only of **branches**, which represent **systems**, and **nodes**, which represent **signals**.



$$V(s) = R_1(s)G_1(s) - R_2(s)G_2(s) + R_3(s)G_3(s);$$

$$C_2(s) = V(s)G_5(s) = R_1(s)G_1(s)G_5(s) - R_2(s)G_2(s)G_5(s) + R_3(s)G_3(s)G_5(s)$$

$$C_3(s) = -V(s)G_6(s) = -R_1(s)G_1(s)G_6(s) + R_2(s)G_2(s)G_6(s) - R_3(s)G_3(s)G_6(s).$$

• Signal-Flow Graphs

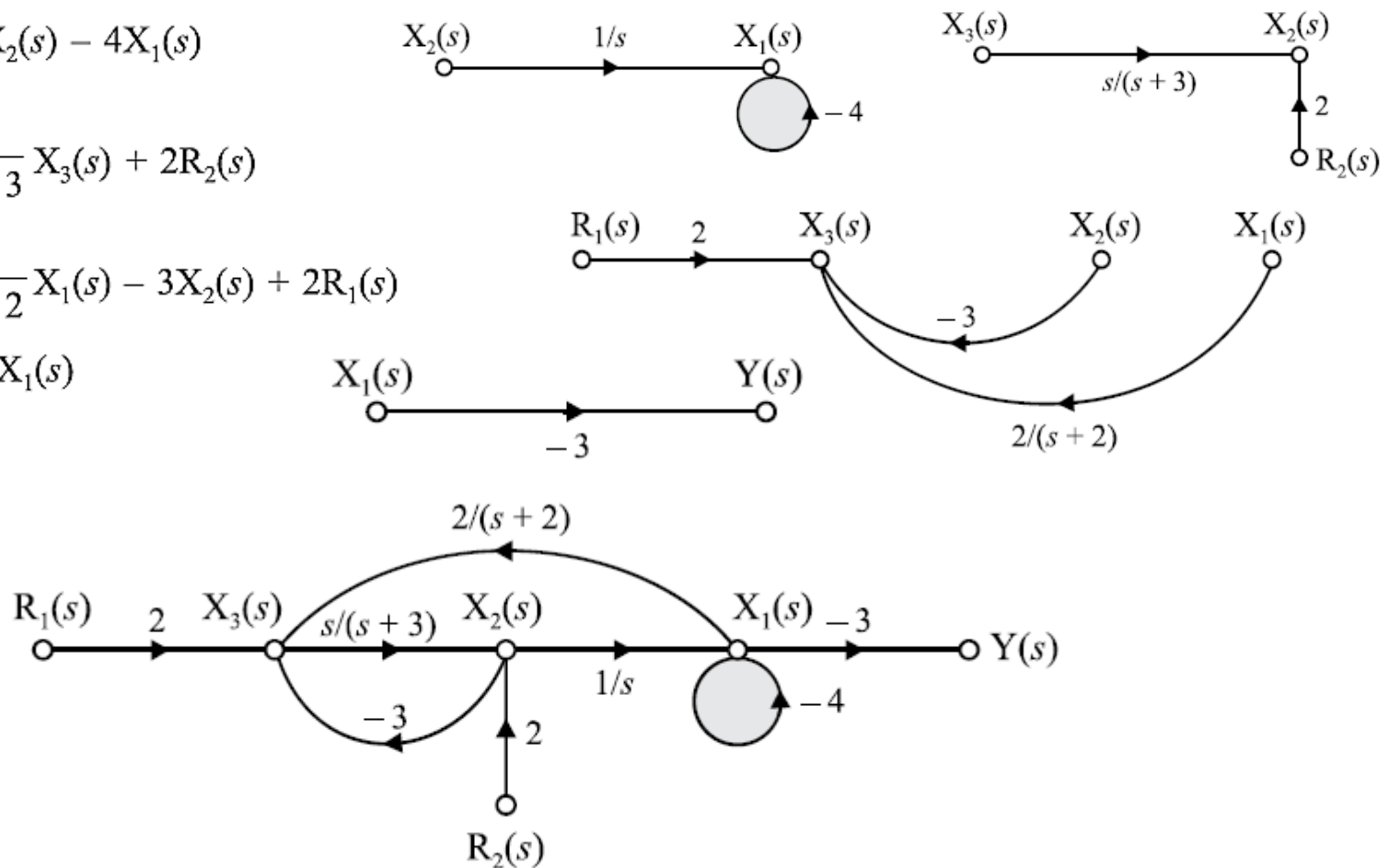
- Let us consider the following Laplace transformed equations to illustrate the construction of the signal flow graph:

$$X_1(s) = \frac{1}{s} X_2(s) - 4X_1(s)$$

$$X_2(s) = \frac{s}{s+3} X_3(s) + 2R_2(s)$$

$$X_3(s) = \frac{2}{s+2} X_1(s) - 3X_2(s) + 2R_1(s)$$

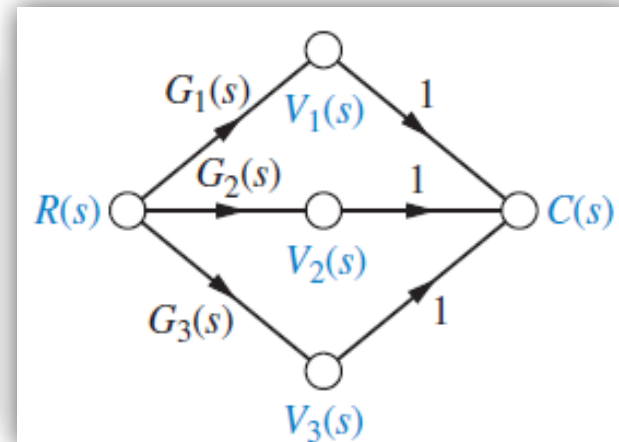
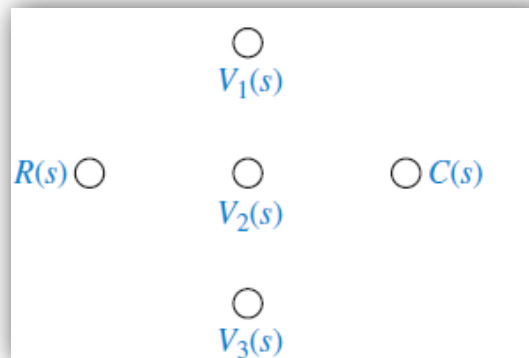
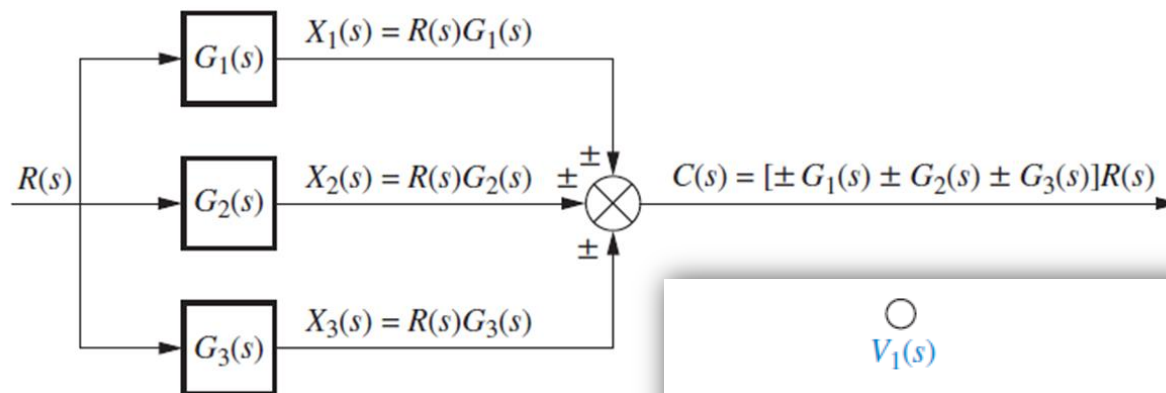
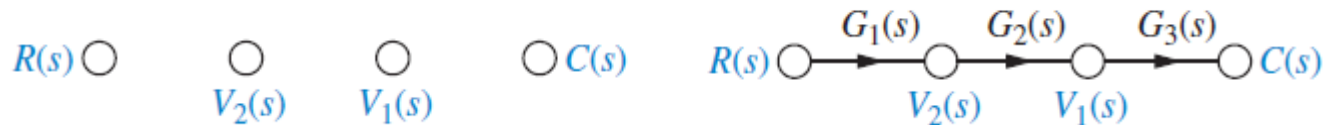
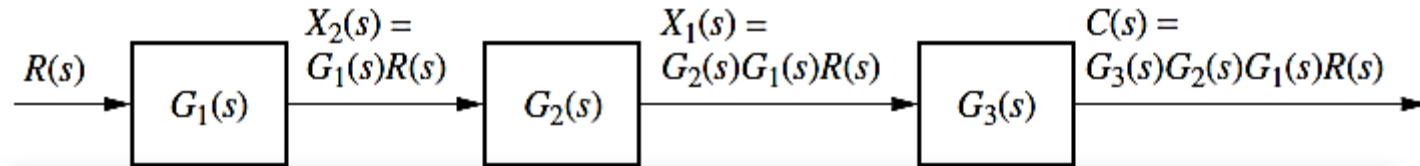
$$Y(s) = -3X_1(s)$$



Complete signal flow graph

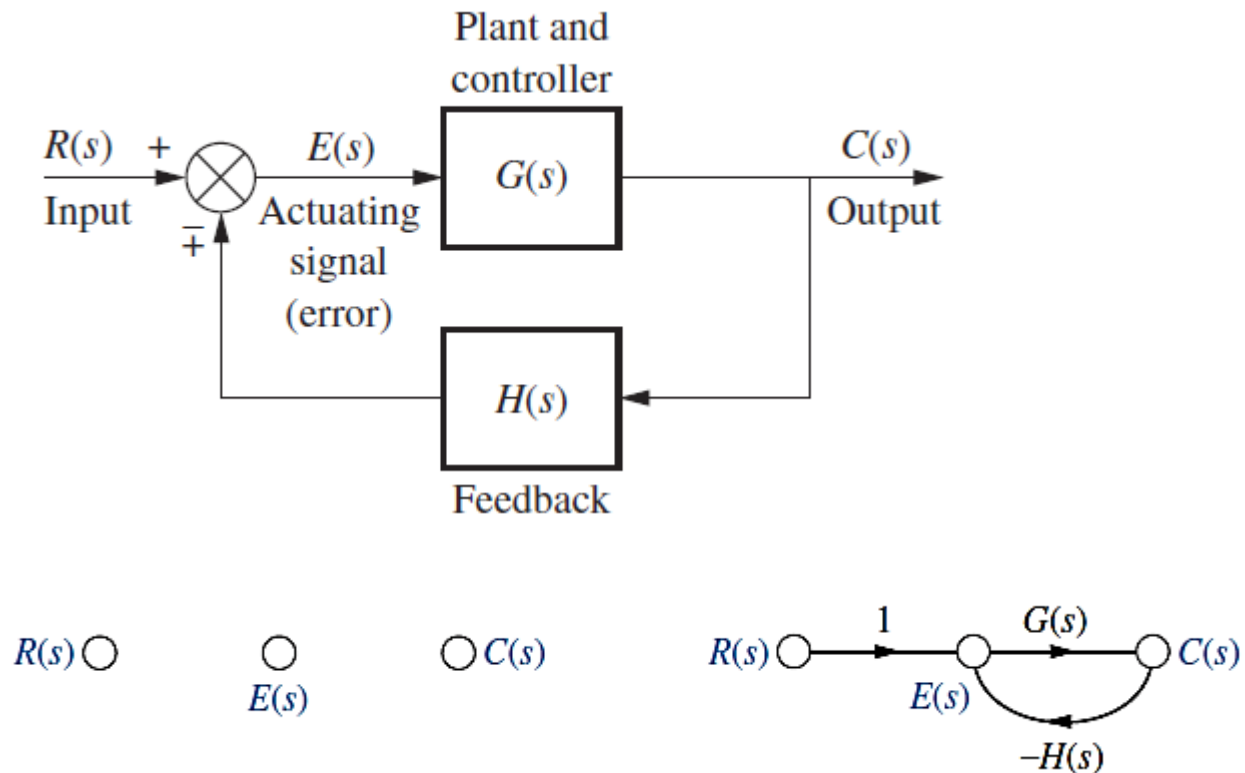
- Signal-Flow Graphs

PROBLEM: Convert the block diagram of Figure to a signal-flow graph.



- Signal-Flow Graphs

PROBLEM: Convert the block diagram of Figure to a signal-flow graph.



PROBLEM: Convert the block diagram of Figure to a signal-flow graph.

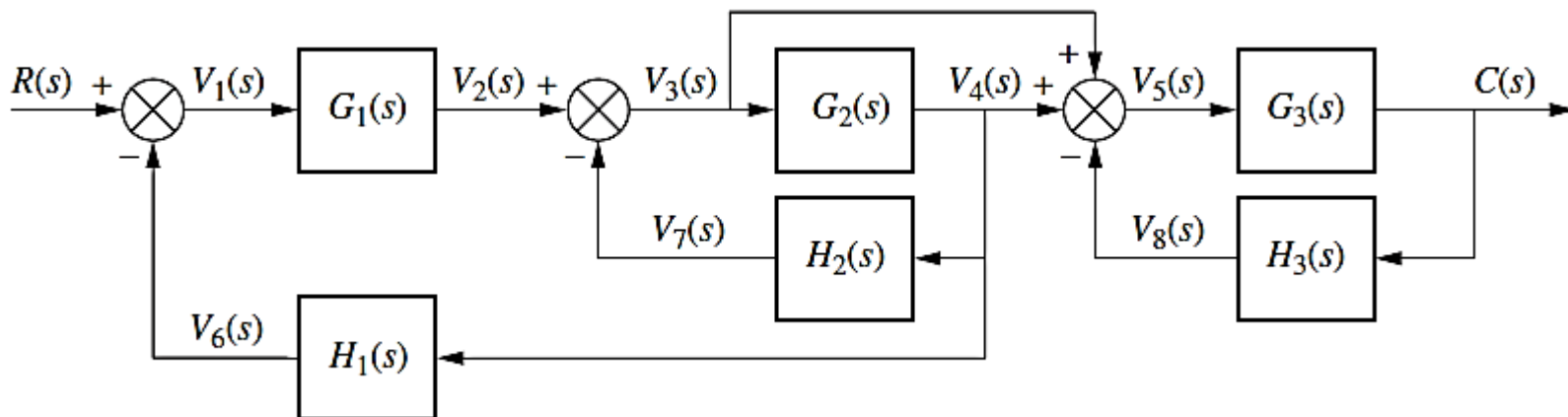
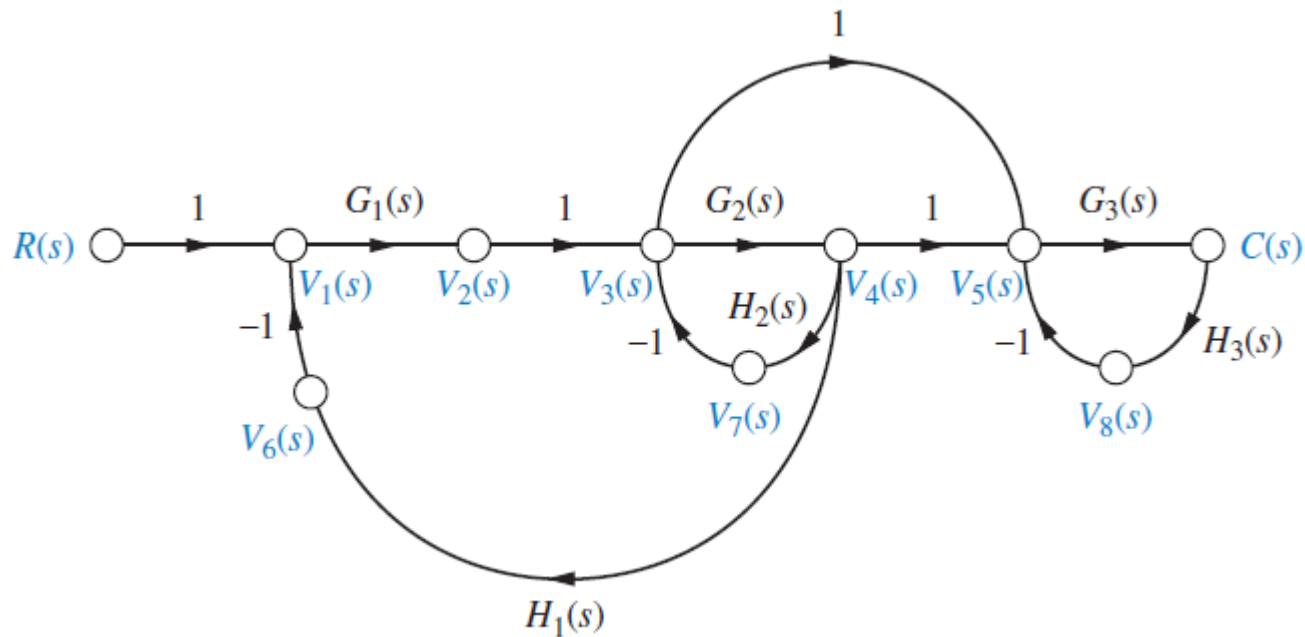
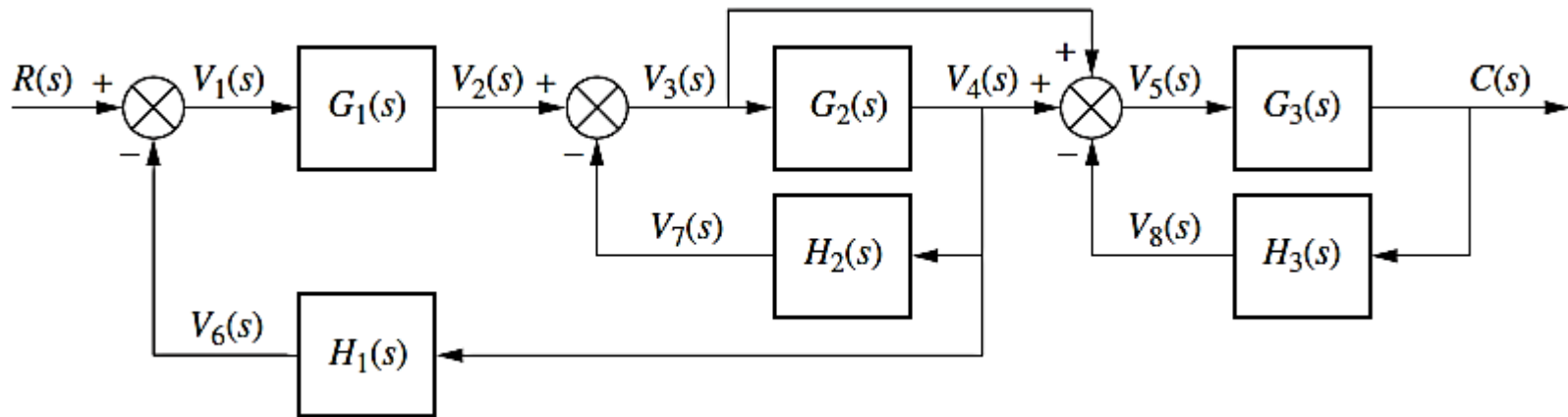
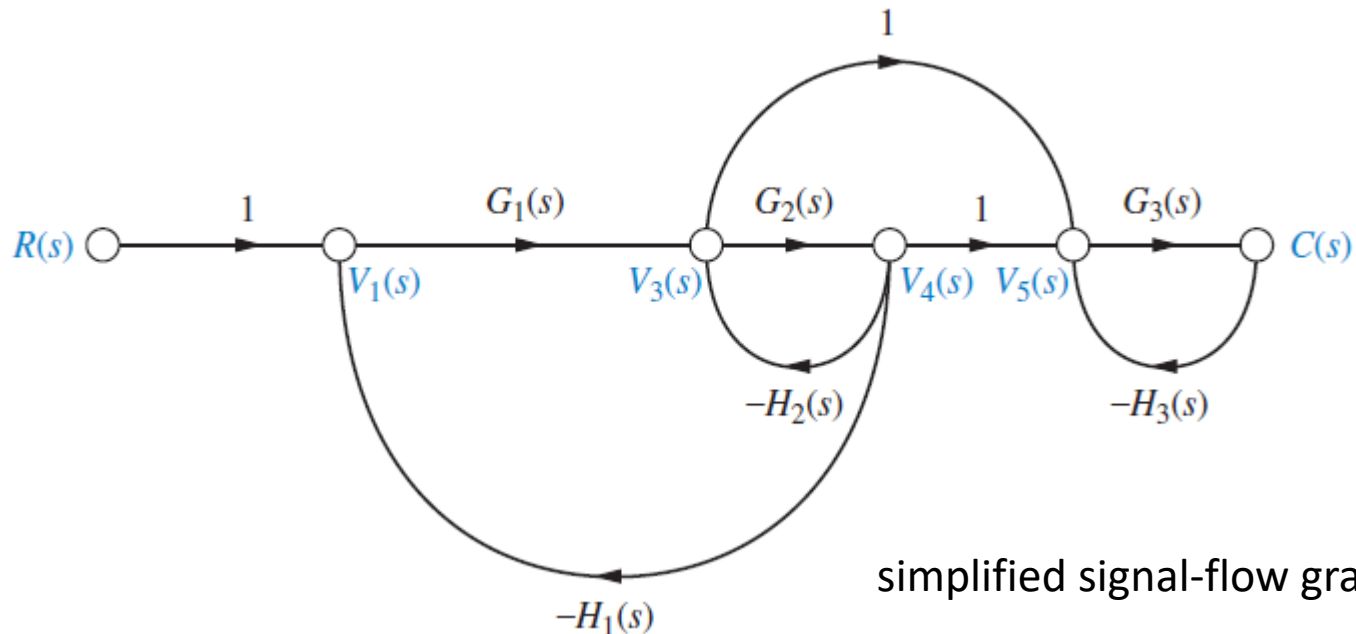
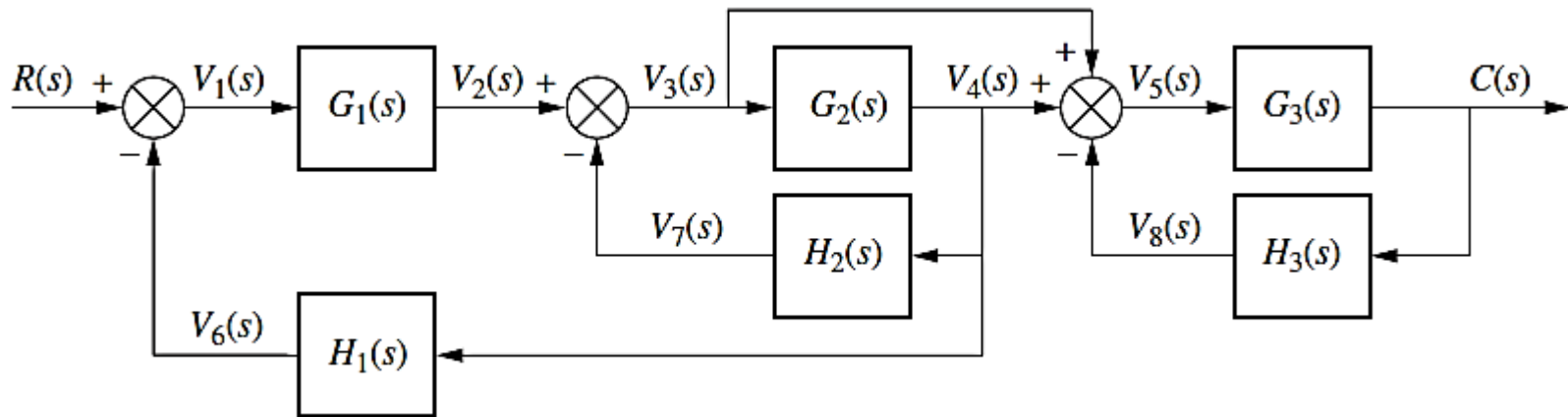


Diagram of a feedforward neural network with two hidden layers. The input layer has nodes $R(s)$ and $C(s)$. The first hidden layer has nodes $V_1(s)$, $V_2(s)$, $V_3(s)$, $V_4(s)$, and $V_5(s)$. The second hidden layer has nodes $V_6(s)$, $V_7(s)$, and $V_8(s)$. All nodes are represented by circles.

PROBLEM: Convert the block diagram of Figure to a signal-flow graph.



PROBLEM: Convert the block diagram of Figure to a signal-flow graph.



simplified signal-flow graph

End of Lectures 5-6

